

1 P3 C4: one above a triangular number

For a partition λ of n , Bulgarian solitaire decreases every positive part by one, discards zeroes, and adds a new part equal to the number of original parts. Write B for this operation, $d_B(\lambda)$ for the least number of moves before the trajectory enters a cycle, and $D_B(n) = \max_{\lambda \vdash n} d_B(\lambda)$. Put $T_j = j(j+1)/2$.

For every integer $k \geq 5$,

$$D_B(T_{k-1} + 1) = (k-1)(k-3).$$

This is a written mathematical proof of both bounds. It uses the general cycle classification already proved in [C1](#) and Lean, and the pile-lifetime descent bound proved in [C2](#). The complete C4 argument is not formalized in Lean.

The pile-count pattern method comes from [Griggs and Ho, Sections 3–4](#). Their Theorem 4.5 supplies this extremal family and a lower bound; the upper-bound argument below is supplied explicitly, not inferred from their conjecture about arbitrary card counts. No novelty claim is made.

1.1 Lifetimes and the last pattern

Put $n = T_{k-1} + 1$ and let c_i be the number of piles in $B^{i-1}(\lambda)$. The pile born on move i exists at integer times $i+1, \dots, i+c_i$; call its lifetime J_i . Thus

$$c_{i+1} = c_i + 1 - d_i,$$

where d_i is the number of piles dying at time i . In particular, a rise by one permits no death, and a constant consecutive pair permits exactly one death. A block of m pile counts satisfies

$$\sum_{j=0}^{m-1} c_{p+j} \leq n + T_{m-1}.$$

(1)

Indeed, the piles existing at time p contribute at most their total n remaining cards to these counts. The successive new piles contribute at most $m-1, m-2, \dots, 1$ further occurrences.

A pattern of level x and width $L \geq 2$ is

$$(c_p, \dots, c_{p+L}) = (x-1, x, \dots, x, x+1).$$

[C2's retreat argument](#) proves

$$p \leq x(L-1).$$

(2)

It also proves that a level $k - 1$ pattern of width k satisfies

$$p + k \leq n + 1.$$

(3)

Both are statements about arbitrary solitaire trajectories, not just triangular card counts. Their lifetime proofs precede the triangular application in C2. Equation (2)'s abstract induction is additionally [checked in Lean](#).

Choose the earliest index t such that $B^{t-1}(\lambda)$ is cyclic and $(c_t, c_{t+1}) = (k - 1, k)$. Such an index exists: C1 identifies the cycle with a rotating length- k word containing exactly one 1. The count is k in one phase and $k - 1$ in the others. Write d for the first cyclic depth. Then $d \leq t - 1$.

If $t \geq k + 1$, there is either

- a type-I pattern with level k , endpoints $t - k \leq p < q \leq t - 1$, hence width $L \leq k - 1$; or
- a type-II pattern with level $k - 1$, endpoints $t - k + 1 \leq p < q \leq t + 1$, hence width $L \leq k$.

Here is a direct proof of that dichotomy. Among $J_{t-k}, \dots, J_{t-1}$ at least one is absent at t , since $c_t = k - 1$; the last is present. Let u be the latest absent index. The rise at t allows no death. Hence J_{u+1} survives through $t + 1$, whereas J_u ends before t . The bound $c_{u+1} \leq c_u + 1$ forces

$$c_u = t - u - 1, \quad c_{u+1} = t - u.$$

If $u > t - k$, then $c_u \leq k - 2$, and the rise from that value to $c_{t+1} = k$ contains a type-II pattern: take the last value at most $k - 2$ and the first subsequent value at least k . Otherwise $u = t - k$ and $(c_{t-k}, c_{t-k+1}) = (k - 1, k)$. Any subsequent value at most $k - 2$ before t gives type II; any value at least $k + 1$ gives type I by the same last-crossing construction.

It remains to rule out all those intermediate values lying in $\{k - 1, k\}$. Put $m = t - k$. In the state at time m , let H_a be the number of piles of height at least a . For $1 \leq a \leq k$, every one of the $a - 1$ piles born since time m is still alive at time $m + a - 1$, since its birth size is at least $k - 1$. Therefore

$$H_a = c_{m+a-1} - (a - 1) \in \{k - a, k - a + 1\}.$$

At time $m + k = t$, the $k - 1$ piles $J_{m+1}, \dots, J_{t-1}$ already fill every slot, so no pile from time m has height greater than k . These conjugate-height bounds place the earlier state between the two consecutive staircases. Moreover, $\sum_a H_a = n = T_{k-1} + 1$, so exactly one height exceeds its lower staircase value by one. Conjugating gives the binary-boundary form of C1, hence the state is cyclic. Since $(c_m, c_{m+1}) = (k - 1, k)$, this contradicts minimality of t and proves the dichotomy.

1.2 Upper bound

Set $F = (k - 1)(k - 3)$. If $t \leq k$, then $d \leq k - 1 \leq F$. Otherwise use the dichotomy above.

For type I, applying (1) to its $L + 1$ entries gives

$$n \geq k(L + 1) - T_L.$$

The right side is nondecreasing for integer $L < k$ (its successive difference is $k - L - 1$). If $L \geq k - 3$, it is at least $T_{k-1} + k - 3 > n$, since $k \geq 5$. Hence $L \leq k - 4$, and (2) gives

$$d \leq t - 1 \leq p + k - 1 \leq k(k - 5) + k - 1 = k^2 - 4k - 1 < F.$$

For type II consider its width L .

If $L \leq k - 3$, then (2) and $p \geq t - k + 1$ give

$$d \leq p + k - 2 \leq (k - 1)(k - 4) + k - 2 = F - 1.$$

Width $L = k - 1$ is impossible. The pile born on move p has size $k - 2$, so it dies at $p + k - 2 = q - 1$. The final rise from $c_{q-1} = k - 1$ to $c_q = k$ permits no death there.

Suppose $L = k - 2$. Inspect the state at time $p + 1$, immediately after the first rise. It has $k - 1$ piles, including the new pile of size $k - 2$. At times $p + 1, \dots, p + k - 4$, exactly one pile dies each time; none dies at $p + k - 3$. No pile born on or after move p can die at those times. Thus the old piles have sizes exactly $1, 2, \dots, k - 4$, together with two sizes at least $k - 2$. Their total, including the new pile, is at least

$$T_{k-4} + 3(k - 2) = T_{k-1}.$$

There is just one extra card, and the newly born pile has fixed size $k - 2$. The two remaining old sizes must therefore be $k - 1$ and $k - 2$. The state is exactly

$$(k - 1, k - 2, k - 2, k - 4, k - 5, \dots, 1),$$

which is a cyclic binary boundary by C1. Consequently $d \leq p$, and (2) gives $d \leq (k - 1)(k - 3) = F$.

Finally suppose $L = k$. At time $p + 1$ there are again $k - 1$ piles, including the new pile of size $k - 2$. Exactly one pile dies at each time $p + 1, \dots, p + k - 2$, and none at $p + k - 1$. The new pile dies at $p + k - 2$; no later-born pile dies this early. Thus the old piles have sizes $1, \dots, k - 3$ and one size at least k . The total is at least

$$T_{k-3} + (k - 2) + k = T_{k-1} + 1 = n.$$

Equality forces the state $(k, k - 2, k - 3, \dots, 1)$, again cyclic. Therefore (3) gives

$$d \leq p \leq n + 1 - k \leq F.$$

The last inequality follows from $2(F - (n + 1 - k)) = k^2 - 5k + 2 \geq 2$ for $k \geq 5$. This exhausts the possible widths and proves the upper bound.

1.3 Matching explicit family

Take

$$\lambda_k = (k - 2, k - 2, k - 3, \dots, 2, 2, 1).$$

Equivalently, in k columns its entries are $\lambda_1 = k - 2$, $\lambda_i = k - i$ for $2 \leq i \leq k - 2$, and $(\lambda_{k-1}, \lambda_k) = (2, 1)$. Its sum is $T_{k-1} + 1$.

Use zero-based columns and one-based rows; a card at row i , column j has diagonal index $i + j$. Relative to the full staircase of height $k - 1$, this diagram has one hole on diagonal $k - 1$ at column 0, and two additional cards on diagonal k at columns $k - 2, k - 1$.

Before sorting, every move rotates diagonal w by one column modulo w . Let $0 \leq s < F$ and write $s = a(k - 1) + b$, where $0 \leq a \leq k - 4$ and $0 \leq b \leq k - 2$. The hole is at b ; the two added cards are at

$$(b - a - 2) \bmod k, \quad (b - a - 1) \bmod k.$$

For either expression, if its representative before reduction is nonnegative it is strictly less than b . Otherwise reduction adds k , giving a column at least $b + 2$. Thus neither extra card is above the hole, nor immediately to its right. The diagram has no vertical gap and its heights remain decreasing: away from the hole, adding at most one card per column cannot reverse the staircase's adjacent order; only an extra card immediately right of the hole could do so. Therefore sorting changes nothing for all these times, and induction proves this exact trajectory description.

Every state with $s < F$ still has its hole below the boundary, so C1 shows it is not cyclic. At $s = F$, the unsorted diagram has its hole at column 0 and its extra cards at columns 1, 2. Its first three heights are $k - 2, k - 1, k - 2$; sorting swaps the first two. The result is

$$(k - 1, k - 2, k - 2, k - 4, \dots, 1),$$

which is cyclic. Thus the first cyclic depth of λ_k is exactly F , proving the matching lower bound and the formula.

1.4 Finite cross-check

Run `python3 cagent/p3/c4/check.py` from the repository root. The independent checker exhausts every partition for $k = 5, \dots, 9$, confirms maxima 8, 15, 24, 35, 48, and verifies the displayed witnesses. [Recorded output](#) covers 27,491 partitions. This finite replay supports the audit; the argument above proves the unrestricted formula.